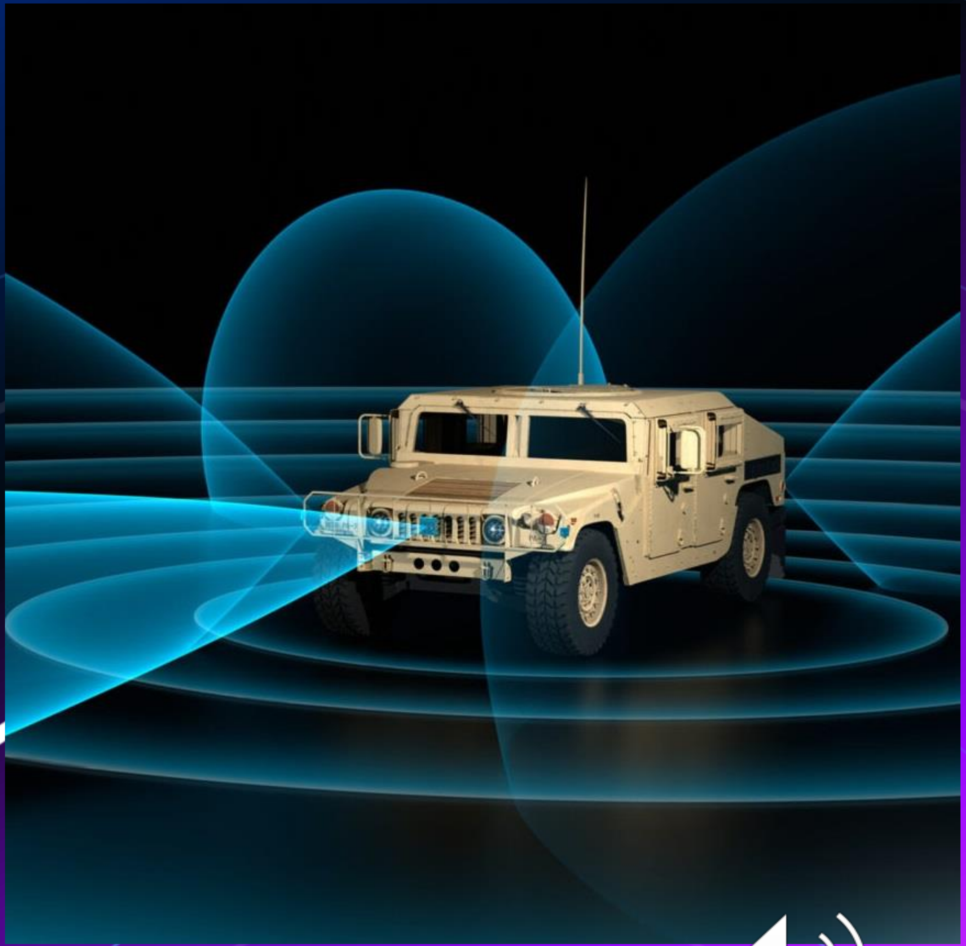
# ADVANCED TACTICAL CONVOY SOLUTIONS

## UNCREWED GROUND VEHICLE (UGV) APPLICATIONS IN MODERN WARFARE

# AGENDA

- Executive Summary
- UGV Mockup
- Mission Summary & GSA Tender Review
- Stated Requirements
- Recommended Additional Requirements
- System-Level Summary
- Design Concept Review
- UGV Selection Process and Justification
- Summary



# EXECUTIVE SUMMARY

- Installs in less than 5 hours
- Overall additional mass < 250kg
- GNSS-Denied tracking ±2m
- 12-hour runtime
- <$120k cost per unit

# EXECUTIVE SUMMARY CONT.

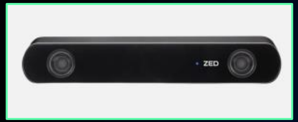
- Supports multi-UGV convoys
- Validated in GNSS-denied scenarios (SIL/HIL)
- Redundant communications paths
- Field power-tested
- Ready for fleet staging

# HMMWV M998 UGV

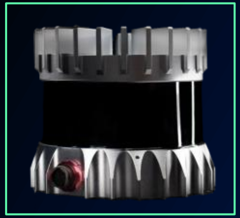
MOCKUP

- Antenna Equipment

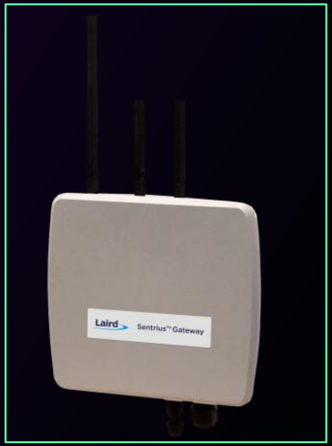
- Sensor Pod




- Stereo Camera



- LIDAR Sensor



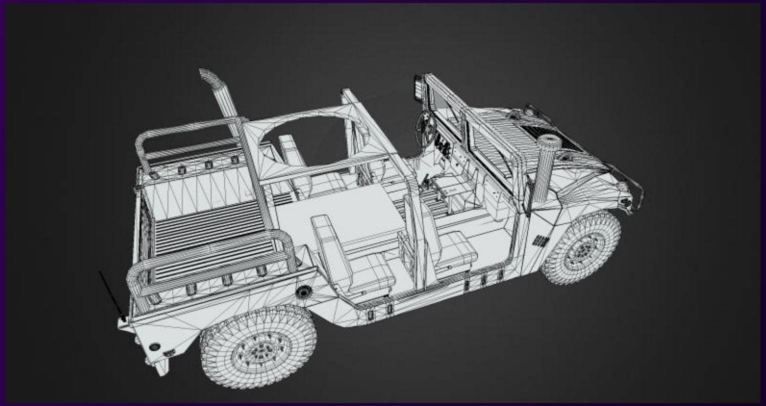

• Communications Equipment

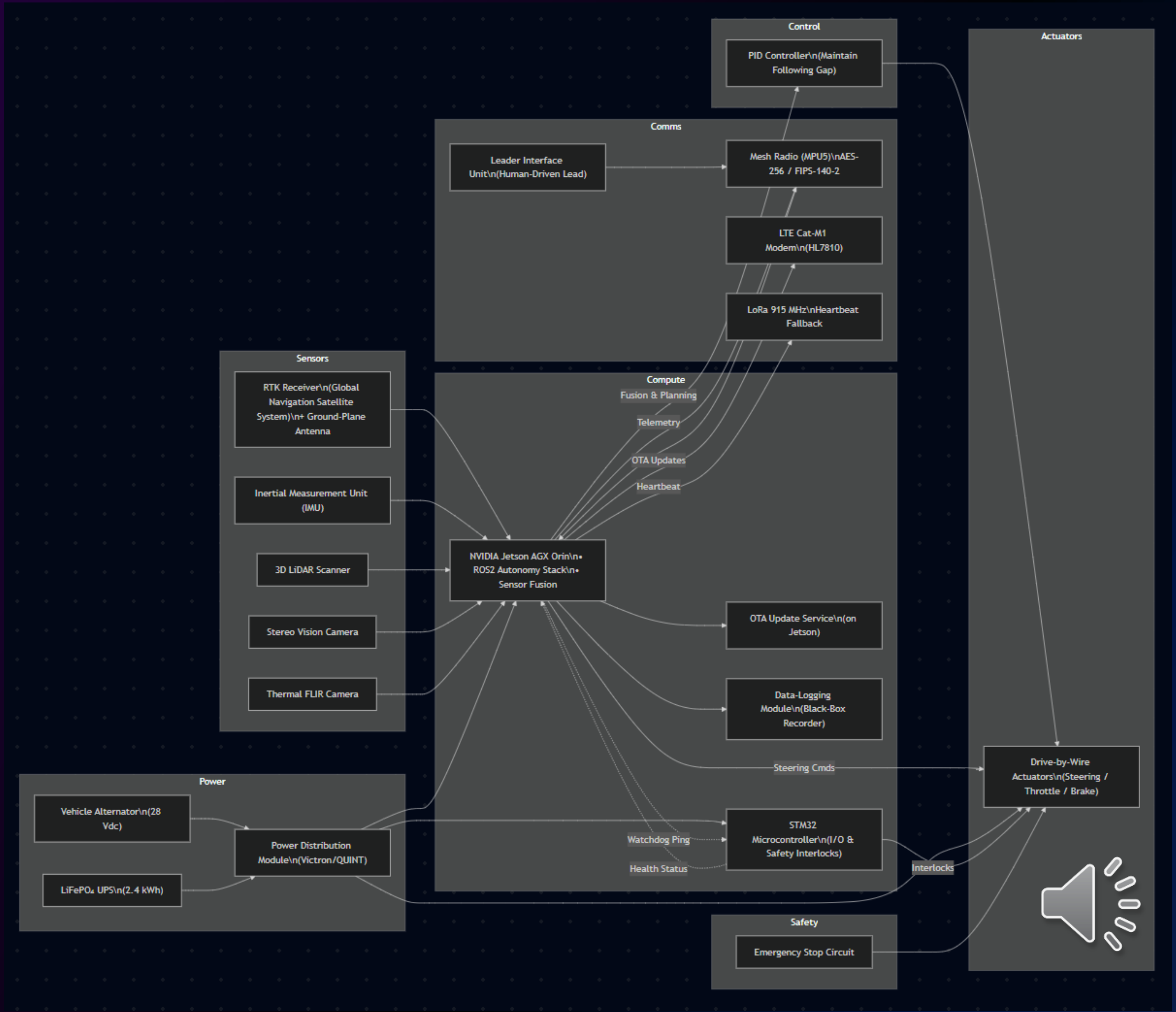
• Computer Equipment


- Remote Operation - Vision


- Remote Operation - Control

# SYSTEM ARCHITECTURE





# MISSION SCENARIO

- Follow lead HMMWV through off-road terrain
- Minimize performance degradation (speed, endurance, terrain compatibility).
- Support ≥500 lbs. payload.
- Maximize autonomy; allow occasional human input.
- Consider future expansion to multiple UGVs per leader vehicle.

# TENDER SPECIFICATIONS

## Minimum Requirements

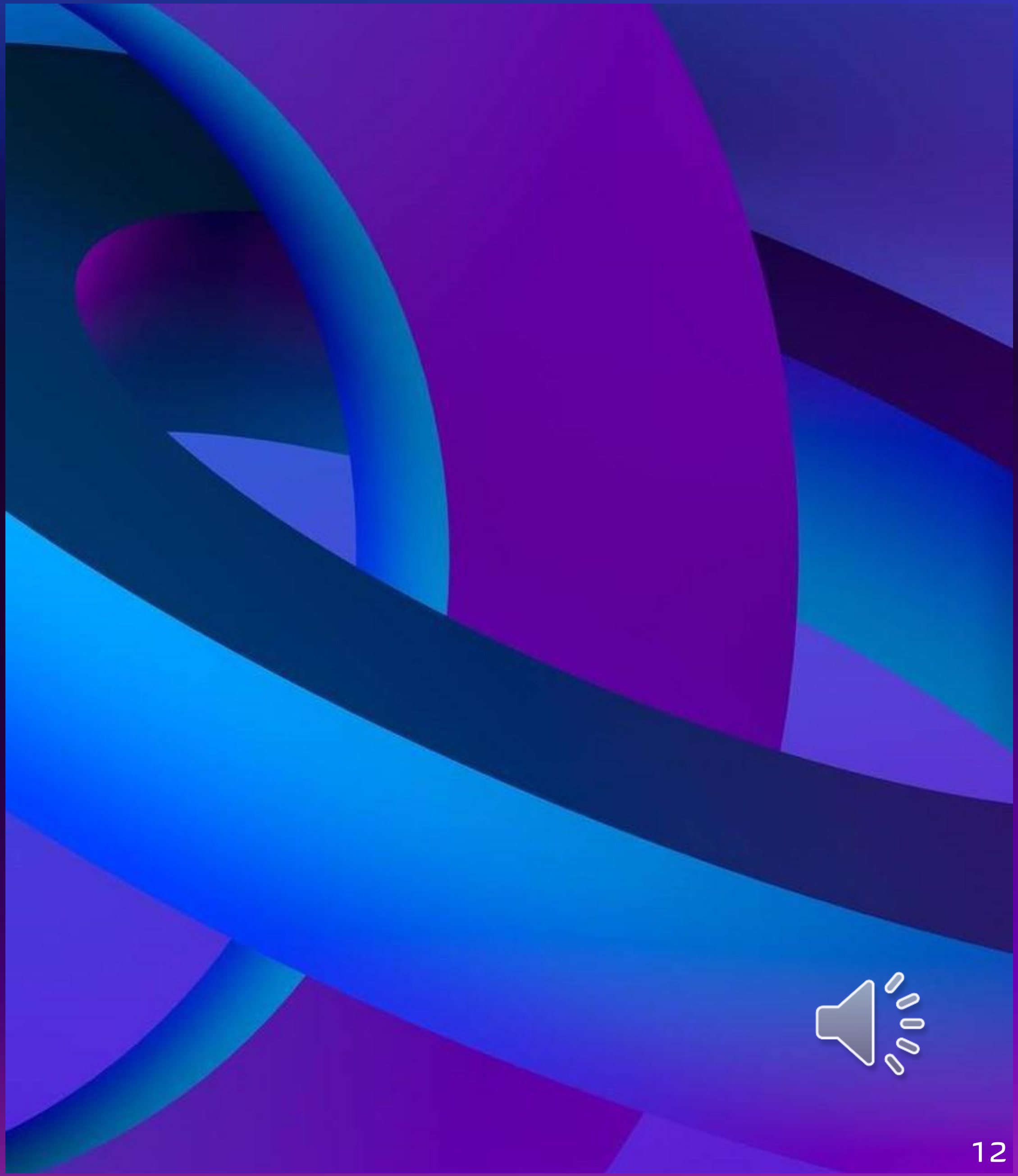
- Follow HMMWV off-road with minimal impact.
- Carry 500 lb payload.
- Operate with no continuous supervision.
- Avoid significant modification to lead vehicle.

## Recommendations

- Allow remote/local override of autonomous functions.
- Navigate in GNSS-denied environments.
- Enable OTA updates.
- Build scalability into initial offering.

SYSTEM

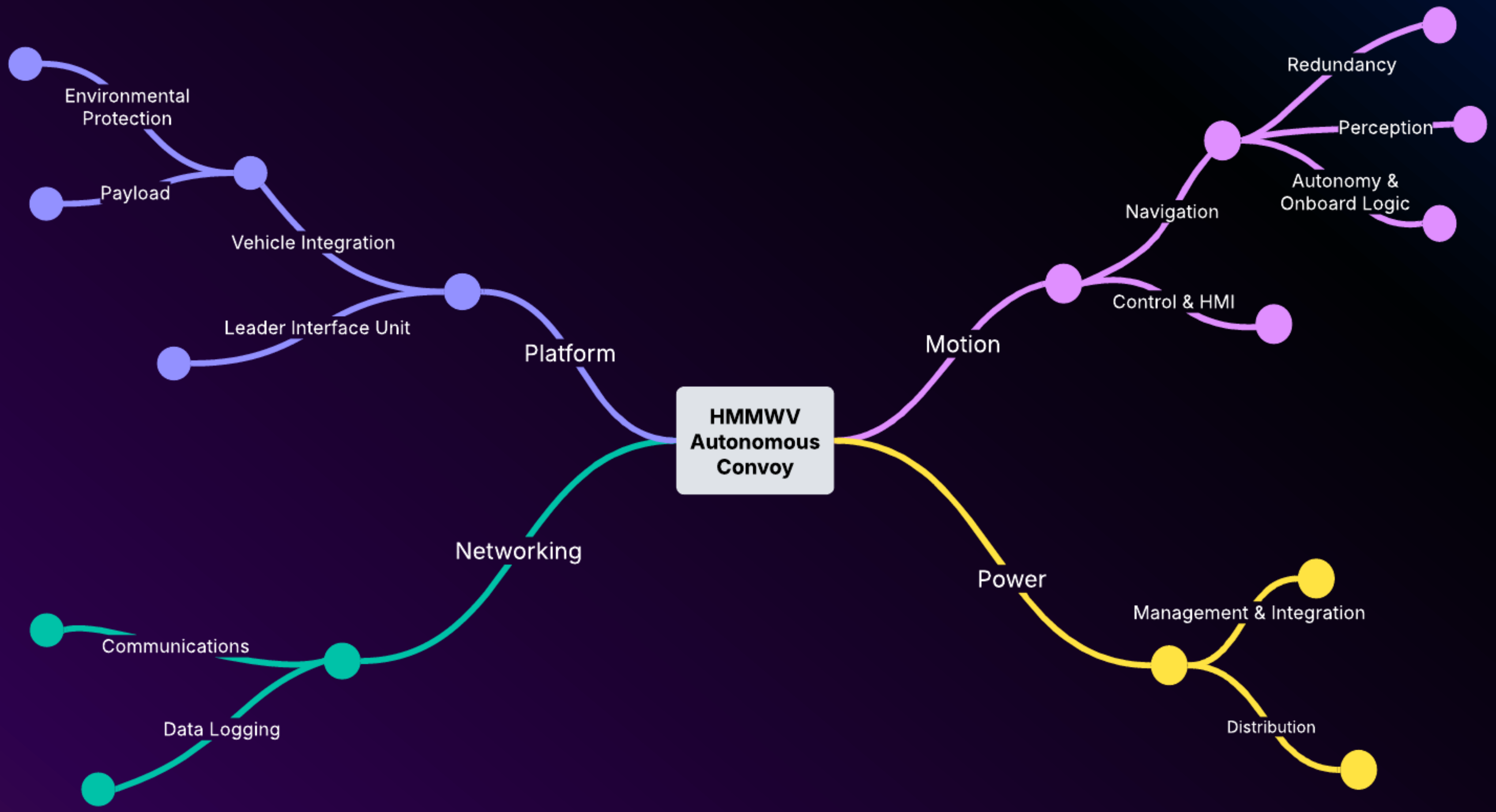
# REQUIREMENTS



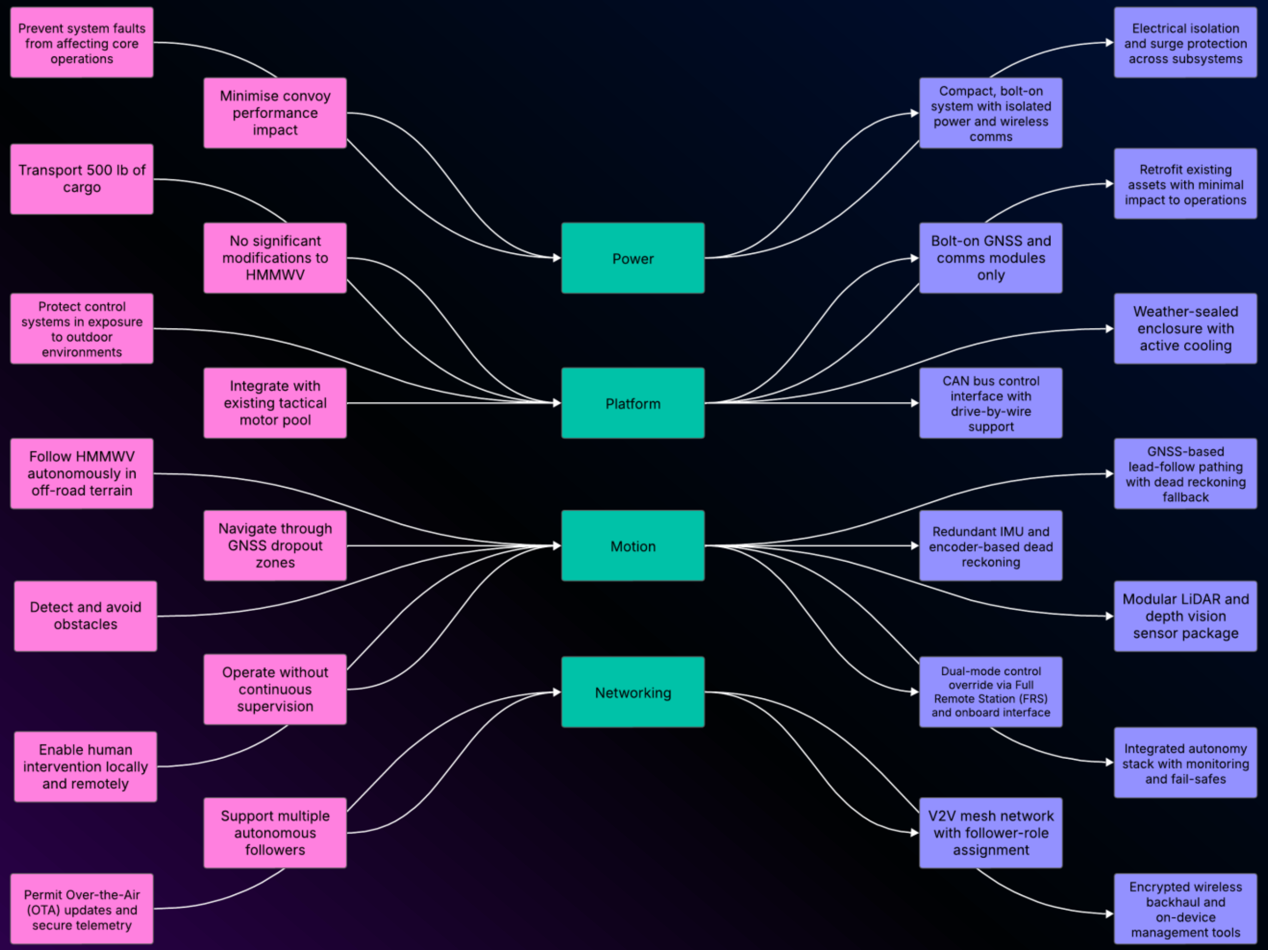
# REQUIREMENTS DEFINITIONS SAMPLE

| Function | Requirement | Difficulty | Risk | Subsystem |
|---|---|---|---|---|
| Navigation | Autonomous Leader-Follower | High | High | Nav Stack |
| Logistics | Mission-Ready Cargo Capability | Medium | Medium | Chassis/Payload |
| Autonomy | Near-Zero Action | Med | High | Autonomy Stack |
| Communications | SAT/BT/WiFi/OTA | Med | Med | Telemetry Stack |
| Control | Intervention Capable | Low | Low | UI Stack |

# SUBSYSTEM AMALGAMATION



# TRACEABILITY MATRIX



# TRADE STUDIES AND ANALYSIS - PERFORMED

- External trade studies performed in relevant sectors of the industry that provide analytical justification for methodologies, as well as valuable lessons learned for future architecture implementations:
  - Verification of heterogeneous multi-agent system using MCMAS
  - *Usability of Perception Sensors to Determine the Obstacles of Unmanned Ground Vehicles Operating in Off-Road Environments*

# DESIGN PROCESS

OVERVIEW

# DESIGN STRAT
# AND DEVE

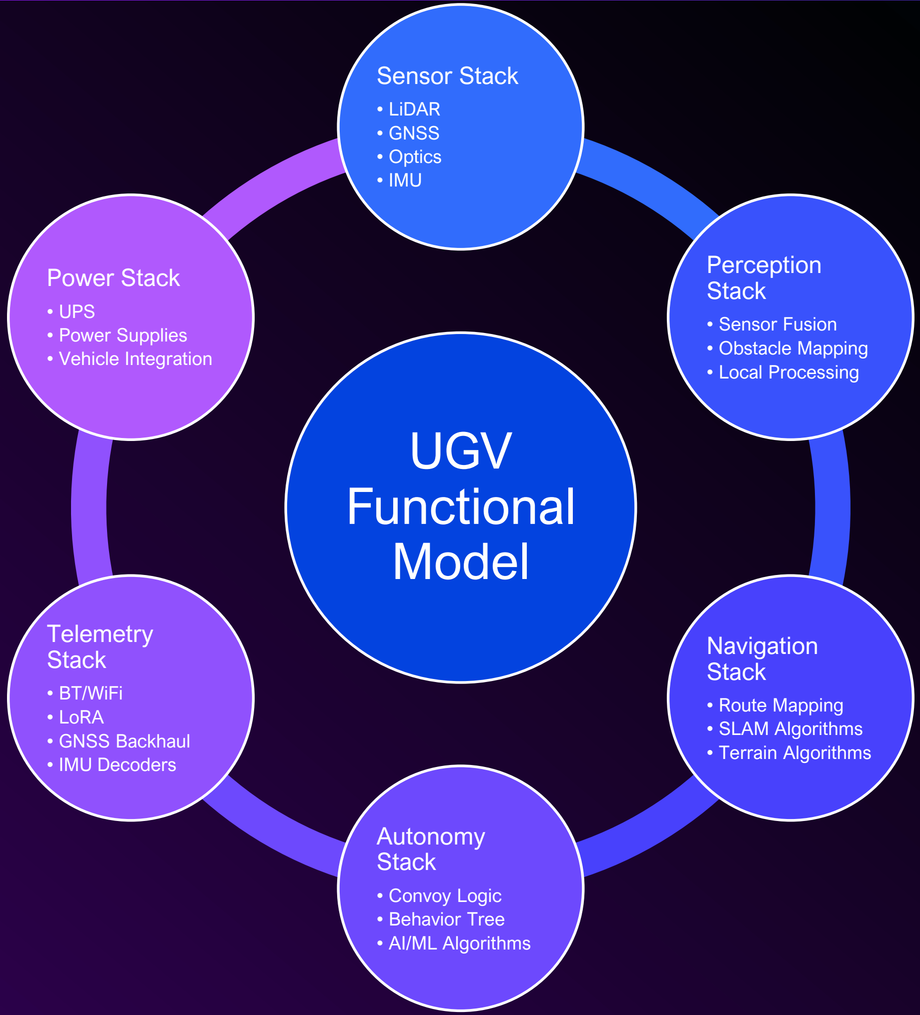
# ASSUMPTIONS

1. Lead HMMWV remains unmodified (minor bolt-on only)
2. COTS and ITAR-compliant hardware
3. Terrain: NATO Class I–III
4. AO: Open terrain, semi-urban, no dense city-navigation
5. No invasive wiring or structural modifications
6. Temperature band: -20°C to +55°C

# CONSTRAINTS

1. Retrofit mass < 250kg
2. Install time <5 hours
3. Runtime >12 hours
4. Size must not exceed tactical vehicle footprint
5. Fully self-contained functionality
6. No hydraulic, vacuum, or engine-tap dependencies

# STACK ARCHITECTURE



# DEPLOYMENT SCHEDULE

### Design Validation
- Months 1-3

### HMMWV Integration
- Months 4-6

### Field Testing on Tactical Motor Pool
- Months 6-9

### Retrofit Kit Production
- Month 10

### Initial Deployment
- Month 12

# RETROFIT ARCHITECTURE SUMMARY

- Non-invasive drive-by-wire control interface
- Steering controlled via non-invasive electric actuator integrated with existing steering system
- Bolt-on GNSS, LiDAR, stereo vision, and V2V radios
- Rugged onboard compute unit (MIL-SPEC compliant)
- Power drawn from host vehicle or auxiliary battery
- No structural modifications required

# PLATFORM COMPATIBILITY/IMPACT

- Compatible with HMMWV, MRAP, JLTV, and DAGOR
- Preserves all original vehicle dimensions
- 250mm choke ring antenna is removable/stowable
- Fully compliant with convoy and transport logistics
- No impact on air, rail, or sea transportability

# VERIFICATION - VALIDATION

- Unit tests on perception, SLAM, and communications systems
- Software-in-the-loop (SIL) and hardware-in-the-loop (HIL) simulations
- Environmental test chambers for sensors
- Range testing of V2V mesh network and GNSS fixed-solution resiliency in combination with dead reckoning IMU decoder

- Convoy testing in real-world terrain (Class I–III off-road routes, DoD Facility Combat Driver Training Courses)
- Operator feedback from simulated remote override trials
- Field scenarios: Obstacle avoidance, GNSS dropout, failed leader-detection
- Acceptance criteria: Spacing accuracy, obstacle miss-rates, uptime percentage

# STRATEGIC ADVANTAGES TO A RETROFIT SOLUTION

- Preserves investment in existing DoD vehicles
- Supports scalability to multi-UGV convoys
- Rapid integration using existing vehicle interfaces
- Designed for GNSS-degraded environments with OTA support
- Cost-effective autonomy solution without platform redesign

# UGV SELECTION

OVERVIEW

*"THE UGV SHOULD STRIVE TO LIMIT THE CONVOY'S SPEED, ENDURANCE, AND TERRAIN CAPABILITY AS LITTLE AS POSSIBLE AS COMPARED TO THE HMMWV"*

# LEAD VEHICLE ANALYSIS

Figure 1

*Lead Vehicle: M998 A0 Series*

*Sourced from Military.com*

- Diesel
- 4 Wheeled
- 6.2 Liter V8
- 3 Speed Automatic
- Driveshaft/Propeller Shaft Rear Axle with 2 U-Joints
- Suspension: independent 4x4

- Weight: 5,220 to 5,920 lb curb weight, 7,020 to 8,520 gross weight
- Length: 15 ft (4.57 m), wheelbase 10 ft 10 in (3.30 m)
- Width: 7 ft 1 in (2.16 m)
- Height: 6 ft (1.83 m), reducible to 4 ft 6 in (1.37 m)

# UGV OPTION ANALYSIS

Figure 2

*HMMWV - M998 A0 Series*

*Sourced from Military.com*

- L x W x H: 15’ x 7’ x 6’
- 70 mph top speed
- 350-mile range
- Diesel (multi-fuel)
- 2500 lbs. max payload

Figure 3

*JLTV – Joint Light Tactical Vehicle*

*Sourced from Military.com*

- L x W x H: 20.5’ x 8’ x 8.5’
- 70 mph top speed
- 300-mile range
- Diesel (multi-fuel)
- 5100 lbs. max payload

Figure 4

*HDT – Hunter WOLF*

*Sourced from HDTglobal.com*

- L x W x H: 5’ x 4.5’ x 4’
- 10 mph top speed
- 200-mile range
- Electric power
- 2200 lbs. max payload

# PROPOSED FOLLOWER UGV, POWER TRAIN, SIZE & WEIGHT

Figure 4

*UGV follower: M998 A0 Series*

*Sourced from Getty Images*

- Uniform Size, Speed, and Range
- Diesel Engine, multi-fuel capable
- 4-wheel drive, off-road ready
- 3-speed automatic
- GVW 7700 lbs.
- L x W x H – 15’ x 7’ x 6’
- UGV Retrofit Component Weight – <500 lbs.

## SUMMARY

- Provision of a resilient, modular system architecture
- Designed to exceed stakeholder needs by providing a future-proof UGV operational model
- Positioned for expansion into multi-follower convoy configurations
- Bolt-on retrofit to existing tactical motor pool assets

## NEXT STEPS…

- Collaborate on defining scope and priorities
- Identify subsystem configuration preferences, finalize design, and begin controlled system-level simulations
- Prepare for the first integrated convoy field test in a suitable environment

## QUESTIONS?

# THANK YOU

Joel Garcia, Trey Hardin, Christopher Sperduto

Embry-Riddle Aeronautical University

UNSY-603: UAS Operational Configuration

May 25, 2025

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